

Jyothir Anurag Vasu Chinnaboina

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Education

University of Wisconsin–Madison – *B.S. in Computer Science and Data Science*

Madison, WI

Awards: Dean's List (4 semesters)

Expected May 2026

Coursework: Operating Systems, Algorithms, AI/ML, Statistics, Database Management, Data Structures

GPA: 3.7/4.0

Experience

Mainframe Intern (Part-time) – *Wisconsin Department of Administration*

Jun 2025 – Present

- Deployed an IaC automation platform (Ansible, Terraform) adopted department-wide, streamlining 15+ batch job workflows across state mainframe systems.
- Reduced manual triage time by 90% with a Python pipeline parsing system logs into structured datasets, flagging remediation status across 5+ teams.
- Eliminated 4+ hours/week of manual reporting via a Python orchestrator with YAML config, JSON Schema validation, and Jinja2 templating.

Software Developer Intern (Part-time) – *Vijay Rekha Life Sciences*

Aug 2025 – Present

Sole Developer · vijayrekha.chinnaboina.com

- Shipped a production lab system (FastAPI, PostgreSQL) processing 300+ samples daily with automated validation, abnormality flagging, and Alembic-managed schema migrations.
- Accelerated clinical report validation by 85% using a batch processing pipeline (Python, pdfplumber, regex) that extracts and cross-validates patient data from 100+ PDF reports across multiple formats.

Computer Science Tutor – *Undergraduate Learning Center, UW–Madison*

Sep 2024 – Aug 2025

- Mentored 600+ students across 8 CS courses and evaluated 160+ tutor candidates across 3 hiring cycles, designing technical interview questions for the Undergraduate Learning Center.

ERP Integration Intern – *United States Pharmacopeia*

Jun – Aug 2024

- Improved data retrieval time by 70% with an Oracle ERP analytics tool (Python, SQL, pandas) using optimized SQL queries joining 5 tables to surface order hold trends.
- Automated quarterly stakeholder reports, previously 1 week of manual work, with an ETL pipeline (Matplotlib, Seaborn) rendering visualizations into Word, PPTX, and PDF for executive distribution.

Projects

ApplyOps | *Python, asyncio, Unix Sockets, ChromaDB, SQLite WAL, pytest*

Present

Personal pipeline infrastructure · github.com/Anurax1321/ApplyOps

- Crafted an asyncio Python daemon over Unix domain sockets serving 5 vector collections at sub-second latency from 30s cold.
- Validated 3 concurrent runs to zero SQLite errors via WAL + rebuild locks; 26 pytest tests across retrieval, cache, and daemon fallback.
- Engineered graceful daemon fallback so the Claude Code hook degrades to direct subprocess (~30s) without breaking the conversation.

Office Hours Queue | *Python, Go/Gin, PostgreSQL, Redis, Docker*

Sep 2025 – Present

Co-Founder · *Advisor: Prof. Louis Oliphant (CS)* · ohq.cs.wisc.edu

- Launched a distributed platform serving 500+ students reducing wait 1–3hr to 15min via SSE + Redis pub/sub on 6 containerized microservices.
- Designed a Redis pub/sub pipeline handling concurrent queue updates across 8 Gunicorn/gevent workers with keepalive heartbeats and ZSET-based tiered ordering synced to PostgreSQL for consistency.
- Migrated the Python/Flask backend to Go/Gin to eliminate GIL bottlenecks, leveraging goroutines and channels for concurrent SSE streaming and Redis pub/sub handling across real-time updates.
- Enabled one-command deployments for 6 services with Docker Compose and Traefik reverse proxy, automating CI/CD via GitHub Actions and Portainer webhooks with zero-downtime rollouts.

E-Beam 3D Printer Control System | *Python, Node.js, Express, GCP*

Jan – Aug 2025

Morgridge Institute for Research (Prof. Kevin Eliceiri)

- Architected a concurrent, multi-threaded Python control system integrating 6+ hardware subsystems over Serial, Modbus RTU, and custom binary protocols with fault-tolerant exponential backoff and graceful degradation.
- Implemented a custom network protocol driver for the G9SP safety controller parsing 199-byte binary packets with bit-field decoding for 13 interlock channels, checksum validation, and exponential backoff on link failures.

Skills

Languages: Python, Go, Java, JavaScript, TypeScript, C, SQL, Bash

Infrastructure & DevOps: Linux, Docker, Docker Compose, Terraform, Ansible, Git, GitHub Actions, Traefik, Portainer, Gunicorn

Frameworks & Tools: Flask, FastAPI, Pydantic, Uvicorn, Gin, Node.js/Express, asyncio, Pytest

Databases & Cloud: PostgreSQL, Redis, ChromaDB, MySQL, SQLite, Google Cloud Platform

Concepts: Multi-threaded Programming, Fault-Tolerant Design, Concurrent Safety, Graceful Degradation, Zero-Downtime Deployment, Infrastructure as Code, Network Protocol Design, CI/CD